

ERAY YAPAĞCI

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EDUCATION

Bachelor of Computer Science, Bilkent University

CGPA: 3.57

Achieved 362nd place out of over 2 million participants in the Higher Education Institutions Exam (YKS, Y-SAY).

Expected Graduation: September 2026

Comprehensive Scholarship Recipient

EXPERIENCE

Research Assistant — BILSEN (Bilkent University Software Engineering and Data Analytics Research Group), Bilkent University, Ankara, Turkey

September 2024 - Present

- Collaborated within the BILSEN research group under the supervision of Eray Tüzün on automated bug reproduction and refactor acceptance prediction using LLMs.
- First-authored "BugCraft: End-to-End Crash Bug Reproduction Using LLM Agents in Minecraft," a preprint (arXiv:2503.20036) detailing a framework using LLM agents to automatically reproduce crash bugs in an end-to-end manner in Minecraft from user reports.

AI Engineer — DataBoss A.Ş., Ankara, Turkey

June 2024 - Present

Python, LangChain, Vespa, EasyOCR, Docker, PyTorch

- Designed and implemented an automatic document chunking system utilizing contextual retrieval methods for semantic segmentation, enabling offline RAG across various document formats.
- Developed an advanced PDF-to-PDF translation tool integrating layout detection models, OCR technology, and LLMs to accurately preserve document structures during translation.
- Developed an expert chatbot based on a leading academic framework, utilizing Ministry of Education documents to deliver highly knowledgeable, context-aware answers with verifiable source citations.

Backend Intern — Gürış Teknoloji A.Ş., Ankara, Turkey

March 2024 - May 2024

C#, .NET, AngularJS

- Developed the backend for a company-wide job application system and integrated it with the frontend, enhancing the recruitment process efficiency.

Full-stack Intern — Hop, Ankara, Turkey

August 2023

React.js, Redux, C#, .NET, OpenAI API

- Learned React.js and developed a project utilizing ChatGPT and Google Maps APIs to summarize reviews of selected locations.
- Tested the Hop mobile application, documented 10 bugs, and contributed to fixing 3 critical issues, improving app stability.

Backend Intern — Logo Yazılım, Remote

July 2023

C#, ASP.NET, Swagger, Jira

- Gained proficiency in C# and ASP.NET technologies; enhanced an internal API with comprehensive Swagger documentation for Diva Retail software.

PROJECTS

Bilkent AI Math Benchmark

Benchmark Site

- Developed a benchmark consisting of 44 high-quality, olympiad-level math problems to evaluate frontier AI mathematical reasoning skills.
- Additionally, created a framework using LLMs to synthetically generate more challenging math problems, specifically targeting questions the generating LLM itself cannot solve, ensuring continued benchmark relevance.

Rubber Duck AI

Python, LangChain, OpenAI API, Neo4j, Vespa, Flask, React.js

- Awarded 1st Place in the Amazon University Engagement Programme for developing an innovative LLM-powered learning platform with adaptive question generation.
- Implemented key features including AI-driven knowledge graph construction, dynamic subgraph identification, interactive feedback mechanisms, and hybrid retrieval systems.

EasyOCR Optimized

GitHub Repository

Python, PyTorch, TensorRT, ONNX

- Enhanced the EasyOCR library and optimized CRAFT inference to achieve up to 3x faster inference on large images.
- Added support for TensorRT and ONNX compilation, improving model inference speed by 20%.

EXTRACURRICULAR ACTIVITIES

- 1st Place, ODTÜ Data Days Datathon.
- Member of the 14th term of QNB FinansUp, a career club by QNB Finansbank, where I developed fundamental project management skills.
- **Disaster Relief Coordinator:** Initiated and led efforts during the 2023 Turkey earthquakes to track aid collection across Ankara, reaching over 200,000 individuals and managing communication with over 11,000 WhatsApp group members.
- Founding Member of Bilkent Natural Disasters Club; Active Member of Bilkent IEEE, and Bilkent Mathematics Society.
- As a Board Member of the Bilkent AI Society, collaborated with Global Turks AI (a Silicon Valley-based think tank) to co-organize events featuring Bilkent alumni from Silicon Valley discussing AI advancements.
- Contributed to organizing the Bilkent IEEE Career Fair 2023 and Bilkent University Integration Bee in 2023 and 2024.
- **Content Creator on Instagram:** @dailymathquestions (original math questions) and @dailymathtricks (solutions and math tricks) — over 10,000 followers and 220,000 views collectively.